



Ownership structure and corporate governance in Chinese stock companies

Xiaonian Xu, Ph.D.^a, Yan Wang, Ph.D.^{b,*}

^a*Head of Research, China International Capital Corporation, #6 Fu Xing Men Wai Avenue, Everbright Building, 17F, Beijing 100045, People's Republic of China*

^b*Senior Economist, The World Bank, 1818 H Street NW, G4-170, Washington, DC 20433, USA*

Abstract

This study investigates whether ownership structure significantly affects the performance of publicly listed companies in China within the framework of corporate governance. A typical listed stock company in China has a mixed ownership structure with three predominant groups of shareholders—the state, legal persons (institutions), and individuals—each holding approximately 30% of the stock. Ownership is heavily concentrated. The five largest shareholders accounted for 58% of the outstanding shares in 1995, compared with 57.8% in the Czech Republic, 79% in Germany, and 33% in Japan. Empirical analysis shows that the mix and concentration of stock ownership do indeed significantly affect a company's performance. First, there is a positive and significant correlation between ownership concentration and profitability. Second, the firm's profitability is positively correlated with the fraction of legal person shares, but it is either negatively correlated or uncorrelated with the fractions of state shares and tradable A-shares held mostly by individuals. Third, labor productivity tends to decline as the proportion of state shares increases. These results suggest the importance of large institutional shareholders in corporate governance, the inefficiency of state ownership, and potential problems in an overly dispersed ownership structure. © 1999 Elsevier Science Inc. All rights reserved.

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Restructuring the state-owned enterprises (SOEs) has been the key to the success or failure of China's economic reform. Two competing approaches have been proposed: the market approach and the ownership approach. The first approach is based on a belief that if the markets for products, factors of production, and corporate control are created and function well, efficiency improvements of SOEs can be achieved without dramatic changes in ownership. Proponents of the second approach argue that private ownership is a necessary condition for enterprise efficiency. Considering the two alternative views, China has adopted a reform strategy

* Corresponding author. Tel.: (202) 458-1411; fax: (202) 676-9810; *E-mail address:* ywang2@worldbank.org (Y. Wang)

Table 1

Ownership concentration of large, nonfinancial corporations—Percentage of outstanding shares owned by the largest five shareholders

	United States	Japan	Germany	Czech Rep., 1995	China (listed companies, $n = 316$) 1995
Mean (%)	25.4	33.1	79.2	57.8	58.1
Median (%)	20.9	29.7	98.5		58.3
Standard deviation	16.0	13.8	31.7	17.5	15.8
Minimum (%)	1.3	10.9	5.0		3.1
Maximum (%)	87.1	85.0	100.0		94.4
Mean firm size: total assets (millions of \$, 1980)	3,505	1,835	3,483		162.4*
Mean firm size: market value of equity (millions of \$: 1980)	1,287	811	1,497		148.9*

Source: Dittus and Prowse (1996), p. 24, and Prowse (1998). The Czech Republic column is based on Claessens et al. (1996), Table 1. The column on China is calculated by the authors and is not directly comparable with the other three columns.

* These values are in millions of 1995 dollars, obtained by dividing the current RMB value by the exchange rate of \$1 = 8.3 yuan.

that gives priorities to fostering markets and nurturing institutional changes, whereas in Russia and Eastern Europe, radical ownership reforms were put in place at an early stage of reform.

In light of the financial crises in East Asia, it is timely and crucial to study issues related to corporate governance, generally defined as institutional arrangements and mechanisms through which outside investors in the firm control the insiders of the firm to ensure returns on their investment. It is well known that there are two different models in corporate governance, the Anglo-American model and the German-Japanese model (see Aoki & Kim, 1995; Prowse, 1994 & 1998). For transition economies, important issues can be addressed: how corporate governance mechanisms have changed after privatization (or corporatization); whether outside investors' interests are protected by law; and to what extent these changes have led to improvements in corporate performance.

This study investigates whether ownership structure significantly affects the performance of publicly listed companies in China within the framework of corporate governance. With publicly listed stock, one can quantify the ownership mix and concentration, which makes it possible to study this issue. Over 800 companies were listed on the two national stock exchanges in 1998. These companies are typically owned by five groups of shareholders: the state, legal persons (institutions), tradable A-share holders (mostly individuals), employees, and foreign investors. The first three groups are the main shareholders, each controlling roughly 30% of the outstanding shares by the end of 1995. Employees of the companies and foreign investors together hold less than 10%. Ownership is heavily concentrated: the five largest shareholders accounted for 58% of the outstanding shares, compared with 57.8% in the Czech Republic, 79% in Germany, and 33% in Japan. See Table 1 for a comparison.¹

¹ For comparative studies, see Annex A in Xu and Wang (1997), Prowse (1994 & 1998), Aoki and Kim (1995), Frydman et al. (1996), and Claessens et al. (1996), among others.

Publicly listed companies, however, represent only a small subset of China's enterprises—a clean and perhaps better performed group of enterprises that were chosen to be listed on the two stock exchanges. These companies started more or less on an equal basis, because they undertook the same restructuring process required by China's Securities Regulatory Commission (CSRC) before their initial public offering. Clearly they are not representative of all enterprises in China, state or nonstate. Therefore the results of our analysis must be treated with caution.

Industrial countries have relied on different mechanisms for corporate governance because of differences in their legal and regulatory frameworks and levels of market development. In the Anglo-American model, share ownership is dispersed, and unsatisfactory performance is sanctioned by shareholders selling shares or by hostile takeovers. In the German-Japanese model, core investors own significant stakes. Concentration of ownership provides core investors with both the incentive and the ability to monitor and control the management. Enterprise cross-shareholding is another distinct feature (See Tables 1 and 2 for comparisons). The first approach can be viewed as the *market discipline* approach and the second as the *ownership control* approach. Indeed, there has been heated debate on the benefits and drawbacks of the two approaches.

Theoretically, both approaches can find their roots in the literature. Fama (1980), for example, argued that if a firm is viewed as a set of contracts, ownership of the firm is an irrelevant concept. A properly functioning managerial labor market may discipline managers and solve incentive problems caused by the separation between ownership and control. Hart (1983) pointed out that competition in the product market reduces managerial slack and thus provides another disciplinary mechanism. Jensen and Ruback (1983) emphasized the role of the market for corporate control. Martin and McConnell (1991) found that the takeover market has re-

Table 2

A comparison of ownership of common stock—Percentage of outstanding shares owned

Shareholders	United States	Japan	Germany	Czech Rep., 1995	China (listed companies, 1995)
All corporations	44.5	72.9	64.0	45.5	28.7
Financial institutions	30.4	48.0	22.0		
Banks	0	18.9	10.0	15.5*	
Insurance companies	4.6	19.6	—		
Pension funds	20.1	9.5	12.0		
Other	5.7	—	—	30**	
Nonfinancial corporations	14.1	24.9	42.0		
Individuals	50.2	22.4	17.0	49***	31.5
Foreign investors	5.4	4.0	14.0	3.4	6.1
Government	0	0.7	5.0	3.2	30.9

Source: U.S. Federal Reserve Flow of Funds, Japanese Flow of Funds, Deutsche Bundesbank Monthly Report. Also in Dittus and Prowse (1996). The Czech Republic column is based on Claessens et al. (1996). The column on China is calculated by the authors based on information on listed companies in 1995. The last two columns are not directly comparable with the other three.

* Bank-sponsored investment funds.

** Nonbank-sponsored investment funds.

*** Includes both individuals and local strategic investors.

stricted, nonvalue maximizing behavior of top corporate managers. However, economists argue that ownership matters because it affects, at least to some extent, the working of the markets. For instance, Grossman and Hart (1980) showed that if a firm's ownership is widely dispersed, no shareholder has adequate incentives to monitor the management closely because the gain from a takeover for any individual shareholder is too small to cover the monitoring cost. Shleifer and Vishny (1986) developed a model to demonstrate that a certain degree of ownership concentration is desired for the takeover market to work more effectively. Large shareholders are able to capture a chunk of the gains from monitoring and are likely to supply it at levels that would be otherwise impossible to reach in diffusely held firms.

Empirical studies so far have presented mixed results regarding the debate on the market versus ownership approaches. Many studies, including Lin et al. (1998) and Lin (1996), emphasized the role of market competition in improving enterprise performance. Groves et al. (1994 & 1995), Jefferson et al. (1992), the World Bank (1996), and Broadman (1995) found that China's partial reform has contributed to gains in output per work and total factor productivity. However, both the superior performance of township and village enterprises over SOEs and the much faster growth of the private sector are frequently cited as evidence of the ownership approach being favored (e.g., Jin & Qian, 1998; Che & Qian, 1998; Qian, 1996; Svejnar, 1990; Woo et al., 1994).

Results are also inconclusive with respect to the U.S. economy. Demsetz and Lehn (1985) found no significant correlation between ownership concentration and profit rates for 511 large corporations. Morck et al. (1988) reported a piecewise linear relationship of Tobin's Q with board member ownership for 371 Fortune 500 firms.² Holderness and Sheehan (1988) analyzed 114 publicly listed corporations in which a majority of shareholder owned at least 50.1% of the common stock. Tobin's Q is higher if most owners are corporations, whereas Tobin's Q and the accounting profit rates are significantly lower for firms with individual majority owners. McConnell and Servaes (1990) found for a sample of more than 1,000 firms that Tobin's Q is positively correlated with the fraction of shares owned by institutional investors. These studies along with others seem to suggest that: (1) there is a positive correlation between share holdings of large investors and a firm's performance, and (2) institutional investors appear to be more effective in monitoring a firm's performance than individual shareholders.

The studies by Grossman and Hart (1980) and by Shleifer and Vishny (1986) provide the theoretical foundation for this paper. We use firm level data from over 300 stock companies from 1993 through 1995. We begin with a descriptive analysis of the ownership structure and corporate governance, followed by regressions linking performance and ownership mix and concentration.

1. Ownership and corporate governance: a description

China's corporatization process started from ground zero in the mid 1980s and developed rapidly in the 1990s. The first stock companies appeared in 1984, but formal trading did not

² Tobin's Q is in general defined as the ratio of the market value to the replacement value of the firm, which can be measured as the market value of equity and debts over replacement value of net fixed assets and inventory.

start until the early 1990s. The Shanghai Stock Exchange (SHSE) was inaugurated in December 1990, and the Shenzhen Stock Exchange (SZSE) opened in April 1991. The number of listed companies, trading volume, and total market capitalization has increased drastically since then. The total number of firms listed increased from 183 in 1993 to over 800 in 1998. Total market capitalization reached US\$206 billion at the end of 1997 (IFC, 1998), accounting for 20% of China's gross domestic product and 10% of emerging market capitalization.³ Readers should not be misled by the figures, however, when estimating the size of the Chinese stock market. Not all shares are publicly tradable. Shares are classified as domestic (A-shares) and foreign (B-, H-, and N-shares) according to the holder's residency. Among the four types of A shares (as defined below), only the A-shares held mostly by individuals are allowed to be traded in the open market.

There are four types of A-shares. The *state shares* are held by the central government, local governments, or solely state-owned enterprises. The government recently declared that the ultimate owner of state shares is the State Council. State shares are not allowed to be publicly traded, but are transferable to domestic institutions on approval of CSRC. By the end of 1995, the state has a controlling stake in 35% of SHSE-listed firms and in 22% of SZSE-listed firms. The *legal person shares* are owned by domestic institutions including stock companies, financial institutions other than banks, and SOEs that have at least one nonstate owner.⁴ Securities firms, trust and investment companies, finance companies, mutual funds, and insurance companies constitute the bulk of non-bank financial institutions. Sales of legal person shares to foreign investors were allowed until it was suspended in May 1996. In 1995, 24% of SHSE-listed companies had legal person shareholders holding more than 50% of outstanding shares, and the same figure is 25% at SZSE. The *tradable A-shares* are held and traded mostly by individuals (some by domestic institutions). There is no restriction on the number of shares traded, nor on minimum holding periods for individuals. It is required, however, that tradable A-shares account for no less than 25% of total outstanding shares when a company makes its initial public offering. This is the only type of equity that is traded among domestic investors at the two exchanges. The volumes reported by the stock exchanges are thus entirely the trading of tradable A-shares. The *employee shares* are offered to workers and managers of a listed company, usually at a substantial discount. These share offerings are designed more as a benefit to employees than as an incentive scheme. Employee shares are registered under the title of the labor union of the company, which also represents shareholding employees by exercise their rights. After a holding period of 6 to 12 months, the company may file with CSRC for allowing its employees to sell the shares in the open market. Only 5% of SHSE-listed companies had employee owners in 1995, but 88% of SZSE companies had employee owners.

B-shares are available exclusively to foreign investors and some authorized domestic securities firms. *H-shares* and *N-shares* are listed on the Hong Kong Stock Exchange and New York Stock Exchange, respectively. In 1995, 24% of companies listed on SHSE have offered

³ See World Bank (1995) and IFC (1998) for a more detailed review and analysis of capital markets.

⁴ The legal person shares here should be distinguished from those shares traded on two automated price quotation systems in Beijing: STAQ and NETS. Companies listed on STAQ and NETS cannot be listed on the SHSE or SZSE. In other words, cross-listing is not allowed.

Table 3

Ownership structure of SHSE- and SZSE-listed companies* as of the end of year (%)

		FST state shares	FLP legal person	FTA individuals	FEM employees	FBS foreigners
SHSE	1993	35.3	31.0	25.5	0.9	7.1
		(28.6)	(28.7)	(19.3)	(2.6)	(13.2)
	1994	34.6	29.7	28.0	0.8	6.9
		(27.5)	(28.8)	(19.8)	(3.0)	(13.7)
	1995	34.2	27.4	28.5	0.4	6.3
		(26.9)	(27.1)	(19.5)	(1.9)	(13.5)
SZSE	1993	32.1	27.7	28.8	4.3	7.2
		(27.3)	(23.4)	(14.7)	(5.4)	(16.8)
	1994	28.2	33.5	32.9	2.0	3.4
		(26.5)	(26.0)	(13.9)	(4.4)	(7.8)
	1995	27.7	29.9	34.5	0.4	6.0
		(25.3)	(23.7)	(14.2)	(2.1)	(12.9)

* Cross-firm averages with standard deviations in parentheses.

B-shares or a combination of the three foreign shares, and 25% at SZSE. This percentage has increased in the recent years as Chinese government encouraged companies to be listed on overseas stock markets.

A typical listed Chinese stock company has a mixed ownership structure with three predominant groups of shareholders—the state, legal persons, and domestic individual investors—each holding approximately 30% of total outstanding shares at the end of 1995. Many listed companies do not issue employee and foreign shares. If they do, these shares account for less than 10% of total stock. Thus they are not the focus of our study. Table 3 shows the average ownership mix of stock companies listed at the two exchanges from 1993 through 1995. The proportion of state shares appears to have declined slightly over time, and so does the proportion of legal person shares. The fraction of tradable A-shares seems to be on the rise. Note that all the ownership fractions have large standard deviations, indicating large variations of ownership mix across firms. On average, the state ownership is higher for SHSE-listed companies than those listed on SZSE, whereas legal persons and individual shareholders seem to be more important among SZSE-listed companies.

Table 4 reports ownership structure of listed companies in 1995 by sectors, that is, manufacturing, retailing, utility, real estate, and conglomerates as classified by the two stock exchanges. At SHSE, the state holds a large stake in manufacturing and utility companies, whereas legal persons as a group are the largest shareholder of the conglomerates. Tradable A-share holders are the dominant owner group only in the retailing industry, with a average interest of 36%. At SZSE, the state lost its dominant position in all industries. Legal person owners are dominating in retailing and utility industries. A-share holders are the main owners in manufacturing, in real estate, and for the conglomerates.

To study the ownership distribution by firm size, we break down the samples according to the book value of the companies' total assets (Table 5). For the SHSE-listed companies, the average fraction of state shares rises steadily with firm size and exceeds 50% for large firms. This

Table 4
Ownership structure by sector (1995)*

	No. firms	FST	FLP	FTA	FEM	FBS
SHSE						
Manufacturing	107	37.1 (27.5)	24.8 (27.3)	26.8 (18.9)	0.3 (1.7)	7.4 (14.1)
Retailing	33	32.2 (20.5)	25.0 (18.8)	36.4 (14.1)	0.7 (2.7)	2.3 (9.5)
Utility	12	36.3 (29.0)	34.8 (27.4)	21.9 (14.7)	0.0 (0.0)	4.0 (13.7)
Real Estate	9	30.5 (28.5)	29.3 (31.3)	26.1 (29.0)	0.0 (0.0)	9.3 (13.9)
Conglomerate	23	20.0 (28.4)	42.4 (31.7)	34.3 (22.2)	0.5 (2.3)	1.5 (5.7)
SZSE						
Manufacturing	69	30.9 (24.9)	25.5 (23.0)	33.5 (13.9)	0.5 (2.5)	7.5 (14.8)
Retailing	11	18.0 (16.0)	40.0 (15.6)	38.4 (13.6)	0.1 (0.1)	3.0 (9.8)
Utility	13	24.6 (28.7)	36.7 (23.3)	28.4 (14.7)	0.1 (0.1)	8.7 (15.1)
Real Estate	20	27.9 (29.4)	33.7 (27.1)	35.7 (13.4)	0.7 (2.5)	1.9 (4.7)
Conglomerate	14	22.5 (24.3)	32.5 (25.1)	39.6 (15.5)	0.1 (0.1)	4.2 (8.9)

*Cross-firm averages with standard deviations in parentheses. Calculated from *Shanghai Stock Exchange Statistics Annual*, 1995, and *Fact Book of Shenzhen Stock Exchange*, 1995.

probably reflects the official position that the state should remain in control of key industries and important firms. The pattern is less clear, however, among companies listed on SZSE.

To summarize, there seems to be a tendency for the proportion of state shares to fall over time and the fraction of tradable A-shares to rise. The primary cause of this shift may have been that the state prefers cash dividends rather than stock dividends or rights offerings as dividends.⁵

1.1. Composition of the board and the supervisory committee

To assess the corporate governance of Chinese stock companies, we now examine the composition of the board of directors and the supervisory committee. Here we classify members of the board and supervisory committee as representatives of the state, legal person owners, tradable A-share holders, and nonowners by their present and previous employment. Board and committee members who have a full-time job at the local Bureau of State Property Management (BSPM) or local finance bureau, for example, are considered representatives of the state.

⁵ This is because new shares acquired by the state cannot be traded in the open market. For liquidity reasons, the state prefers cash dividends to stock dividends or rights offerings, and so do legal person shareholders.

Table 5
Ownership structure by firm size (1995)*

	No. firms	FST	FLP	FTA	FEM	FBS
SHSE						
Small (<500 mn)	77	28.11 (24.1)	31.4 (26.0)	36.1 (17.7)	0.2 (1.4)	2.2 (8.7)
Medium 1 (500 mn–1 bn)	43	31.9 (28.4)	26.3 (27.6)	32.1 (22.6)	0.4 (1.9)	6.7 (14.6)
Medium 2 (1–1.5 bn)	38	35.0 (28.0)	31.7 (28.7)	21.4 (13.9)	0.7 (3.1)	9.3 (15.2)
Large (1.5 bn and over)	26	51.7 (24.3)	14.5 (24.4)	14.7 (12.1)	0.2 (0.7)	8.4 (13.5)
SZSE						
Small (<500 mn)	31	32.3 (24.0)	28.7 (23.6)	32.8 (12.2)	0.1 (0.1)	4.5 (11.3)
Medium 1 (500 mn–1 bn)	46	24.1 (23.7)	33.4 (24.9)	38.7 (13.4)	0.1 (0.1)	2.4 (8.1)
Medium 2 (1 bn–1.5 bn)	32	24.3 (26.4)	32.8 (23.9)	32.7 (13.1)	0.4 (1.9)	7.5 (14.5)
Large (1.5 bn and over)	18	35.4 (28.8)	18.2 (16.9)	29.7 (18.7)	1.6 (4.8)	15.0 (17.8)

* Firm size is classified by the book value of their total assets. Cross-firm averages with standard deviations in parentheses. Calculated from *Shanghai Stock Exchange Statistics Annual*, 1995, and *Fact Book of Shenzhen Stock Exchange*, 1995.

Summary statistics on the composition of the boards of 154 companies in 1995 are given in Table 6. A comparison of Table 6 with Table 3 reveals that the board membership is not proportional to the ownership. It is striking to see that individual shareholders have no more than 0.3% of the seats on average, even though they possess approximately one third of the

Table 6
Composition of the board (%)*

	No. firms	State	Legal person	Individual shareholder	Nonowner	Management	Largest shareholder
SHSE							
	83	51.0 (40.0)	39.6 (37.5)	0.2 (1.2)	9.3 (15.6)	48.7 (18.1)	
State dominated	48	74.4 (26.2)	19.5 (23.7)	0.0 (0.0)	6.1 (11.6)		74.3 (24.0)
Legal person dominated	35	19.2 (32.6)	68.4 (32.3)	0.2 (1.2)	12.2 (14.3)		54.4 (26.3)
SZSE							
	71	50.9 (38.3)	39.1 (33.1)	0.3 (1.8)	9.7 (17.4)	50.5 (26.4)	60.5 (34.2)
State dominated	46	71.7 (25.7)	23.9 (23.9)	0.0 (0.0)	4.4 (10.9)	55.1 (24.0)	69.1 (27.7)
Legal person dominated	25	12.5 (25.8)	67.1 (29.4)	0.8 (2.9)	19.6 (22.6)	45.5 (28.9)	45.8 (39.5)

* Cross-firm averages with standard deviations in parentheses. Calculated from 1995 annual reports of listed companies.

Table 7
Composition of the supervisory committee (%)^{*}

	No. firms	State	Legal person	Individual shareholders	Nonowner	Employee
SHSE	83	56.9 (44.3)	32.7 (41.4)	0.0 (0.0)	10.4 (19.8)	77.7 (27.0)
State dominated	48	82.1 (30.5)	11.1 (24.2)	0.0 (0.0)	6.8 (14.3)	84.8 (22.1)
Legal person dominated	35	19.8 (33.8)	64.9 (41.1)	0.0 (0.0)	15.3 (24.6)	65.8 (31.1)
SZSE	71	50.9 (38.3)	39.1 (33.1)	0.3 (1.8)	9.7 (17.4)	50.5 (26.4)
State dominated	46	71.7 (25.7)	23.9 (23.9)	0.0 (0.0)	4.4 (10.9)	55.1 (24.0)
Legal person dominated	25	12.5 (25.8)	67.1 (29.4)	0.8 (2.9)	19.6 (22.6)	45.5 (28.9)

^{*} Cross-firm averages with standard deviations in parentheses. Calculated from 1995 annual reports of listed companies.

total outstanding shares. However, the state is overrepresented on the board: 50% of positions are filled by government officials, which substantially higher than its 30% of average stake in the stock companies. Board membership for management is common for both SHSE and SZSE companies, averaged at 50%.

The composition of the supervisory committee is also characterized by an extremely low presence of individual shareholders, approximately 0.5% for the SZSE-listed companies and zero for the sample companies listed on SHSE (see Table 7). However, employee membership is high: 78% for the SHSE-listed companies and 68% at the SZSE.

Table 8 shows the shareholding of board members and their cash salaries. For the SZSE-listed companies, the average number of shares held by board members is substantially greater for legal person-dominated firms than that for the state-dominated firms. However, board members of legal person-dominated companies appear to receive a higher cash salary than their counterparts in state-controlled companies. The differences in shareholding and salaries of the board members between the state-dominated and legal person-dominated firms are suggestive.⁶ Many board members in the state-dominated firms are still paid according to their administrative rankings, whereas legal person-dominated companies have greater discretion in determining the compensation of their officers.

1.2. Shareholders' behavior and corporate governance

It seems clear that both the state and private institutions rely on their control over the board to influence corporate policies and to protect their interests. The state's interest is rep-

⁶ Data on salaries are limited and inaccurate. Only 33 of the 71 sample companies listed on the SZSE provided such information, and 52 of the 83 SHSE companies did so, even though the disclosure is required by CSRC. As indicated by interviews with a few stock companies, the salary figures in annual reports are not reliable. Reasons for the reluctance in reporting salaries include, first, concerns over possible protest by shareholders, and second, attempts to cover up violations of salary ceilings set by the state.

Table 8

Average shareholding and cash salary of board members*

	Members of the board of directors		Members of the supervisory committee	
	Shareholding	Salary (RMB)	Shareholding	Salary (RMB)
SHSE				
State dominated	4,971 (3,622) [48]	24,242 (14,669) [36]	3,080 (1,985) [48]	17,942 (9,373) [35]
Legal person dominated	5,704 (5,998) [35]	40,612 (16,647) [16]	2,519 (3,478) [35]	26,754 (14,200) [13]
SZSE				
State dominated	11,446 (10,568) [46]	38,680 (65,350) [20]	5,726 (5,605) [46]	
Legal person dominated	18,745 (15,562) [25]	41,446 (29,033) [13]	7,265 (8,092) [25]	

* Cross-firm averages with standard deviations in parentheses and the number of firms in brackets. Calculated from 1995 annual reports of listed companies.

resented by local offices of the BSPM or Ministry of Finance.⁷ The top priority of the BSPM has been stated as “preserving and increasing the value of state properties.”⁸ A series of principal-agent problems may arise from this institutional setting. First, officials from the local BSPM may not have sufficient incentives to preserve and increase the value of state properties. They are civil servants and draw income from the government payroll, which has nothing to do with the performance of the stock companies they oversee. Moreover, these BSPM officials are appointed and paid by the local government. The priorities of the local government do not necessarily coincide with those of the BSPM. Second, it is difficult for the BSPM to verify which decisions made by the management are value increasing and which are value decreasing. The value of state assets is not easy to determine because of the use of book values, which have nothing to do with a company’s profitability. Third, increasing the value of state assets in a firm may lead to conflicts of interest with other shareholders.

The vast majority of China’s individual investors are small shareholders and their interests are not protected. Individual shareholders are in a disadvantageous position because of the lack of proxy voting procedures. For those individuals in the top 10, their holdings are extremely small: normally less than 0.5%. Considering the large stakes held by the state and legal persons, each holding about 30% of the stock, 0.5% held by a single individual is negligible. Almost no individual shareholders are on the board of directors or on the supervisory committee.⁹ It is conceivable that the dispersed individual ownership may give rise to the

⁷ In a few cities such as Shanghai and Shenzhen, government investment companies are established and perform all the functions of BSPM. Little information is available on how these investment companies are organized and operate.

⁸ See, for example, The Bureau of State Property Management. *State Property Management*, Vol. 2, p. 100, (1994), Beijing: The Economics Press.

⁹ There are a few exceptions (approximately five) in which individual shareholders have a board seat. Most of them are the business founders and Hong Kong residents.

classic free-rider problem (Grossman & Hart, 1980). Small investors have neither the incentive nor the capability to monitor managerial performance. The small shareholders' inactivity has also been further worsened by the block holdings of the state and legal persons.

Anecdotal evidence is available to support the hypothesis of free-rider problems. The turnover ratios of the Chinese stock exchanges, for example, are extremely high: over 200% at both exchanges, as compared with 67% in the United State. Because tradable A-shares account for 20% to 30% of total outstanding shares, the effective turnover ratio may range from 700% to 1000%! In other words, the average holding period in China is about 1 to 2 months, whereas it is 18 months in the United States. Apparently, Chinese individual investors are seeking short-term trading profits rather than dividend income or long-term growth. With such a short investment time horizon, it is unlikely that small shareholders are willing and able to monitor the management closely. Small shareholders' participation in the annual shareholder conference is also very low. According to an estimate of CSRC, the average number of shareholders attending annual conferences is approximately 100, whereas the number of shareholders of listed companies ranges from 3,000 to 100,000. The state and legal person owners always send their representatives to the conference with all the expenses covered by employers. Individual investors can go only at their own expense. Very often, shares represented by conference attendants are too low to meet the requirement, and the board must call for an emergency shareholder meeting.

Legal person shareholders in China are not only better motivated, but also better equipped, with power to control and monitor the management.¹⁰ Unlike individual investors, representatives of legal person shareholders are elected to the board of directors and the supervisory committee. Besides their voting power on important issues such as the selection of the management team and dividend policies, they have access to corporate inside information and the right to question chief officers at any time about the operations of the firm. The board also has the privilege of calling for an emergency shareholder meeting, whereas individual shareholders must assemble a coalition that represents at least 10% of the firm's equity to do the same.

2. Ownership structure and a firm's performance: empirical evidence

Using pooled firm-level data for 1993, 1994, and 1995, we first regress performance variables on concentration ratios without distinguishing different ownership mix. The purpose is to investigate the free-rider problem of small investors and the role of large shareholders. Second, we examine effects on a firm's performance of state ownership, legal person ownership, and individual private ownership, respectively. Finally, we look at the impact of both ownership mix and concentration and explore further the roles of legal person shareholders and the state as an owner. Employee and foreign ownership are not addressed in this paper.¹¹

¹⁰ As argued by Shleifer and Vishny (1986 & 1996), large shareholders provide at least a partial solution to the free-rider problem of small investors. They have the incentive to monitor the management even though doing so will also benefit other shareholders.

¹¹ Interested readers may refer to Bailey and Jagtiani (1994), for example, on foreign ownership in the Thai capital market.

Table 9

Ownership concentration: Percentage of shares controlled by the top 10 shareholders (1995)*

	1	2	3	4	5	6	7	8	9	10
SHSE	46.09 (18.4)	6.83 (7.95)	2.88 (3.86)	1.61 (1.98)	1.15 (1.34)	0.84 (1.01)	0.66 (0.80)	0.54 (0.66)	0.47 (0.60)	0.36 (0.50)
SZSE	40.11 (17.4)	8.93 (8.22)	4.08 (3.71)	2.60 (2.37)	1.85 (1.69)	1.32 (1.03)	1.07 (0.84)	0.88 (0.68)	0.78 (0.65)	0.65 (0.68)

* Cross-firm averages with standard deviation in parentheses. Sources: Annual reports of listed companies.

2.1. Ownership concentration and a firm's performance

Let P represent performance variables, and CR be ownership concentration ratios. P is measured by three accounting ratios: the market-to-book value ratio (MBR), return on equity (ROE), and return on asset (ROA). CR is measured by the proportion of shares held by the top 10 shareholders (A10) and the Herfindahl index (HERF) (Table 9 shows the shares controlled by the top 10 shareholders at the two stock exchanges). See the Appendix for a description of variables. If ownership structure does not matter, we find no correlation between P and CR . The null hypothesis is thus stated as follows.

Hypothesis 1 (the irrelevance of ownership concentration). In any regression of P , the coefficient of CR equals zero.

We estimate Eq.(1) to test the hypothesis for the 1995 sample.¹²

$$P = \sum_{i=1}^5 \alpha_i \text{DUM}_i + \phi_1 \text{SALE} + \phi_2 \text{DAR} + \phi_3 \text{GROW} + \phi_4 CR + e, \quad (1)$$

where all Greek letters represent coefficients. The variable e is an error term with a covariance matrix $\text{Cov}(e_j, e_k) = 0$ for $j \neq k$, and $\text{Var}(e_j) \neq \text{Var}(e_k)$. Because the variance of e_j differs across firms, only heteroskedasticity-consistent statistics are reported in this paper. Estimation results of Eq. (1) are given in Table 10.

Hypothesis 1 is rejected decisively because A10 and HERF are significantly different from zero in regressions of MBR.¹³ The correlation of accounting profit rates with the ownership concentration ratios is much weaker. HERF is significant at the 5% level in the ROE equation for the SZSE companies. Neither HERF nor A10 has any explanatory power for ROE of the SHSE companies. Estimation of Eq. (1) with ROA yields qualitatively identical results (not reported here) to those with ROE.

The significant impact of concentration ratios on MBR is in support of the Shleifer and Vishny hypothesis (1986) that large shareholders may help reduce the free-rider problem of small investors and hence are value increasing. This explanation should be taken with extra caution, however. Note that most top 10 shareholders of the Chinese stock companies are state government agencies and legal persons. For example, in the 1995 SZSE sample, only 7 out of 127 companies have individual shareholders in the top 5. There are more individual

¹² We do not have data on top 10 shareholders for the 1993 and 1994 samples.

¹³ Following Demsetz and Lehn (1985), we experimented with logarithm transformation of A10 and HERF in estimating Eq. (1). The results are virtually the same.

Table 10
Ownership concentration and 9 firm's performance (1995)*

	Industry DUM**	SALE	DAR	GROW	A10	HERF	Adjusted R ²	F-statistic
SHSE								
MBR	Yes	−0.00 ^a (−3.92)	1.18 ^b (2.49)	0.02 ^a (2.85)	2.11 ^a (3.10)		0.29	10.4
MBR	Yes	−0.00 ^a (−4.11)	1.11 ^b (2.38)	0.02 ^b (2.13)		1.48 ^b (2.05)	0.27	9.41
ROE	Yes	0.00 ^b (2.37)	−0.45 (−0.10)		0.49 (0.13)		0.05	2.49
ROE	Yes	0.00 ^b (2.37)	−0.62 (−0.13)			1.41 (0.33)	0.05	2.51
SZSE								
MBR	Yes	−0.33 ^a (−3.64)	0.92 ^a (−3.54)	0.33 ^a (3.99)	0.02 ^a (5.86)		0.38	10.5
MBR	Yes	−0.30 ^a (−2.98)	0.76 ^b (2.49)	0.35 ^a (4.11)		1.16 ^b (2.56)	0.26	6.66
ROE	Yes	0.09 ^a (5.22)	−0.10 ^b (−2.59)		0.00 (1.63)		0.25	7.07
ROE	Yes	0.09 ^a (5.09)	−0.12 ^a (−2.94)			0.13 ^b (1.98)	0.27	7.62

* *t*-statistics are in parentheses. ** Industry dummies are significant at the 1% level, but are not reported here.

^a Significant at 1% level.

^b Significant at 5% level.

investors are in the top 10, but the number is fairly small: 1.6 on average. The percentage of shares controlled by individuals in the top 10 is even smaller, with an average of 3.4% of total outstanding shares for the 127 companies. A10 and HERF therefore measure mostly the degree of ownership concentration by the state and legal persons. Accordingly, the results in Table 10 should be interpreted as a positive correlation of MBR with the state and legal person ownership concentration, rather than ownership concentration in general.

2.2. Ownership mix and a firm's performance

We now turn to investigate the effects of ownership mix on a firm's performance. If ownership mix is an irrelevant concept, one would expect ownership fractions to be insignificant in regressions of performance. We first test the irrelevance hypothesis of ownership mix in this subsection, and then investigate how the results change when both mix and concentration variables are included.

Let *F* be ownership fraction variables: the fraction of state shares (FST), legal person shares (FLP), and tradable A-shares held largely by individuals (FTA). The null hypothesis is:

Hypothesis 2 (the irrelevance of ownership mix). In any regressions of *P*, *F* is insignificant.

Hypothesis 2 can be tested by estimating Eq. (2):

$$P = \sum_{i=1}^5 \alpha_i \text{DUM}_i + \phi_1 \text{SALE} + \phi_2 \text{DAR} + \phi_3 \text{GROW} + \phi_4 F + e. \quad (2)$$

We use pooled data from six samples, three for each stock exchange for 1993, 1994, and 1995, respectively. The samples include all listed companies except a few outliers. Outliers are identified in the following way. If an observation falls outside the range of the sample mean plus and minus two times the standard deviation, the observation is dropped. We start with simple regressions for the six samples, 1 year at a time, and then run pooled regressions using the 3-year data as a panel.¹⁴ Results for the pooled regressions are presented in Table 11 for both SHSE and SZSE. The conventional least-square-dummy-variable approach is used in this paper in the pooled regressions because of the characteristics of our data.¹⁵ This approach is widely used in the literature of ownership structure and corporate governance and allows us to improve the efficiency of estimation without incurring heavy cost of technical complexity.

Hypothesis 2 is rejected immediately because the coefficients of three ownership variables differ significantly from zero for the pooled regressions in Table 11. In all the regressions, five industry dummies are included as control variables, and their coefficients are positive and significant at the 1% level. Other controlling variables including operating sales in billion Renminbi (SALE) and the debt-to-asset ration (DAR). Dummies for the years 1994 and 1993 are included, with 1995 as the omitted category. In all regressions, FST has a *negative* coefficient, and it is significant in two of four regressions in Table 11. The firm's performance is *positively* correlated with legal persons' holdings with FLP significant at the 1% level in all four regressions, whether the performance is measured by MBR or ROE or ROA. In contrast, FTA has a significant *negative* effect on the market-to-book ratios for the two stock exchanges. Individual shareholders do not seem to have any significant impact on the profitability of firms as measured by ROE and ROA. The explanatory power of these regressions is fair, with adjusted R^2 ranging from 14% to nearly 40%. We obtain similar results in simple regressions using data 1 year at a time, with somewhat lower t - and R^2 -values.

We then explore the effects of both ownership concentration and mix on performance in Table 12. Indicators of concentration, A5, A10, and HERF are included, and they show strong positive effects. The impacts of ownership mix do not go away. The fraction of state shares has negative coefficients in all regressions, and they are significant in all ROE and ROA regressions.¹⁶ Legal person shares have positive and significant effects on performance, and the t -ratios are higher for ROE and ROA regressions. Individual tradable A-shares have a negative and significant impact in all MBR regressions, but are insignificant in ROE and ROA regressions.

¹⁴ We are grateful to Stijn Claessens for his suggestion of using pooled regressions.

¹⁵ The fixed-effect and random-effect models are not used because first, the former is essentially a dummy-variable classical regression model. The use of the model would reduce the gain in the degree of freedom from panel data if the panel is short. A more serious problem could arise if there is little variation over time in the independent variables. Recall that state and legal person shares are not allowed to be traded at the two exchanges and transfer of these shares among institutions are subject to strict rules. Consequently, the ownership fractions, FST, FLP, and FTA, are fairly stable over time with little variation. Using a fixed-effect model here would mean many zeros for the explanatory variables. Even for the random-effect model, special treatment is needed when some of the independent variables do not change over time.

¹⁶ Because of space limits, only MBR regressions are presented. See Table 3.4.2 of Xu and Wang (1997) for ROE and ROA regressions.

Table 11
Pooled regressions: Ownership mix and a firm's performance, SHSE and SZSE, 1993–95

	Constant	Industry DUM	D93	D94	DUM	SALE	DAR	FST	FLP	FTA	Adjusted R ²	F
Dependent variable equals market-to-book-value ratio ($n = 673$)												
MBR	2.89 (27.7)	No	2.15 ^a (17.98)	0.56 ^a (4.77)	-1.27 ^a (-12.74)			-0.22 (-1.07)			0.23	50.56
MBR	2.57 (24.92)	No	2.16 ^a (18.24)	0.54 ^a (4.70)	-1.27 ^a (-12.71)				0.86 ^a (3.89)		0.25	56.39
MBR	3.36 (22.62)	No	2.05 ^a (16.71)	0.53 ^a (4.72)	-1.16 ^a (-12.45)					-1.87 ^a (-5.65)	0.27	63.19
MBR	No	Yes	2.25 ^a (17.58)	0.60 ^a (5.58)	-1.46 ^a (-13.89)	-0.00 ^a (-5.48)	1.00 ^a (3.07)	-0.13 (-0.62)			0.31	30.88
MBR	No	Yes	2.27 ^a (18.26)	0.59 ^a (5.57)	-1.47 ^a (-13.77)	-0.00 ^a (-5.37)	1.11 ^a (3.51)		0.72 ^a (3.32)		0.32	32.87
MBR	No	Yes	2.12 ^a (16.65)	0.56 ^a (5.48)	-1.32 ^a (-13.77)	-0.00 ^a (-5.96)	0.94 ^a (3.07)			-2.30 ^a (-6.82)	0.37	39.95
Dependent variable equals return to equity ($n = 668$)												
ROE	No	Yes	0.06 ^a (6.75)	0.03 ^a (5.36)	-0.003 (-0.50)	0.00 ^a (3.81)	0.01 (0.87)	-0.03 ^a (-2.73)			0.16	13.42
ROE	No	Yes	0.06 ^a (6.63)	0.03 ^a (5.22)	-0.001 (-0.23)	0.00 ^a (3.80)	0.01 (0.73)		0.03 ^a (2.63)		0.16	13.28
ROE	No	Yes	0.06 ^a (6.47)	0.03 ^a (5.21)	-0.001 (-0.14)	0.00 ^a (3.63)	0.01 (0.35)			0.002 (0.18)	0.14	12.25
Dependent variable equals return to asset ($n = 668$)												
ROA	No	Yes	0.03 ^a (6.26)	0.02 ^a (5.16)	-0.002 (-0.54)	0.00 ^a (3.99)	-0.11 ^a (-12.91)	-0.02 ^a (-2.71)			0.32	32.55
ROA	No	Yes	0.04 ^a (6.21)	0.02 ^a (5.03)	-0.001 (-0.32)	0.00 ^a (4.02)	-0.11 ^a (-13.11)		0.02 ^a (3.17)		0.32	33.09
ROA	No	Yes	0.03 ^a (5.99)	0.02 ^a (4.98)	-0.00 (-0.07)	0.00 ^a (3.69)	-0.12 ^a (-13.62)			-0.01 (-0.90)	0.31	31.28

Heteroskedasticity-consistent t -statistics are in the parentheses. Five industrial dummies are included in most regressions, and their coefficients are all positive and significant at the 1% level.

^a Significant at 1% level.

^b Significant at 5% level.

^c Significant at 10% level.

Table 12
Pooled regressions: Ownership concentration, mix, and a firm's performance SHSE and SZSE, 1995

	Industry	SZSE								Adjusted	
	DUM	DUM	SALE	DAR	GROW	A5	FST	FLP	FTA	R ²	F
Dependent variable equals market-to-book-value ratio ($N = 311$)											
MBR	Yes	−1.47 (−10.89)	−0.00 (−4.52)	1.18 (3.60)	0.00 (2.80)	2.24 (4.47)	−0.34 (−1.27)			0.36	18.11
MBR	Yes	−1.45 (−10.80)	−0.00 (−4.49)	1.18 (3.65)	0.00 (2.86)	1.97 (4.29)		0.45 (1.79)		0.36	18.41
MBR	Yes	−1.32 (−10.29)	−0.00 (−5.0)	1.07 (3.41)	0.00 (2.37)	0.86 (1.57)			−1.61 (−3.54)	0.37	19.80
	Industry	SZSE								Adjusted	
	DUM	DUM	SALE	DAR	GROW	A10	FST	FLP	FTA	R ²	F
MBR	Yes	−1.30 (−10.39)	−0.00 (−4.49)	1.12 (3.34)	0.00 (2.72)	2.18 (4.40)	−0.31 (−1.16)			0.35	17.84
MBR	Yes	−1.29 (−10.35)	−0.00 (−4.46)	1.13 (3.50)	0.00 (2.79)	1.93 (4.30)		0.44 (1.76)		0.36	18.15
MBR	Yes	−1.25 (−10.33)	−0.00 (−5.01)	1.05 (3.33)	0.00 (2.32)	0.72 (1.28)			−1.68 (−3.47)	0.38	19.64
	Industry	SZSE								Adjusted	
	DUM	DUM	SALE	DAR	GROW	HERF	FST	FLP	FTA	R ²	F
MBR	Yes	−1.31 (−10.10)	−0.00 (−4.69)	1.09 (3.31)	0.00 (2.28)	1.82 (2.40)	−0.53 (−1.36)			0.32	15.87
MBR	Yes	−1.30 (−10.04)	−0.00 (−4.62)	1.10 (3.40)	0.00 (2.49)	1.62 (2.86)		0.79 (2.73)		0.34	17.01
MBR	Yes	−1.25 (−10.27)	−0.00 (−5.01)	1.04 (3.23)	0.00 (2.18)	0.36 (0.71)			−1.97 (−5.06)	0.37	19.45

Heteroskedasticity-consistent t -statistics are in the parentheses. Five industrial dummies are included in most regressions, and their coefficients are all positive and significant at the 1% level.

2.3. More on the role of legal person shareholders

Most legal person shareholders have a stake considerably larger than any individual's holding in the sample firms. Large legal person shareholders almost always possess seats on the board of directors and on the supervisory committee. What are their relations with the management: insiders, controllers, or collaborators? McConnell and Servaes (1990) find, by imposing a quadratic functional form, that Tobin's Q increases with insider ownership until it reaches approximately 40% to 50% and declines slowly thereafter. However, Morck et al. (1988) point out that managers respond to two opposing forces. Managers naturally tend to allocate a firm's resources in their own best interests at the expense of outsider shareholders. As management's equity ownership rises, however, their interests become more aligned with those of outside shareholders. The curve that shows the relationship between a firm's value and inside ownership can be downward or upward sloping, depending on which of the forces dominates the other.

Following McConnell and Servaes (1990), we run pooled regressions of the market-to-book value ratio on fractional ownership variables and their squared terms plus a constant in-

Table 13

Pooled regressions: Ownership structure and a firm's performance, nonlinear specifications for FLP

	Industry DUM	D93	D94	SZSE DUM	SALE	DAR	FLP2	FLP	Adjusted R ²	F
MBR	Yes	2.25 (17.94)	0.56 (5.44)	−1.34 (−12.44)	−0.00 (−5.58)	0.99 (3.20)	4.29 (3.75)	−2.46 (−3.21)	0.35	33.18
ROE	Yes	0.06 (6.63)	0.03 (5.18)	−0.00 (−0.04)	0.00 (3.79)	0.01 (0.65)	0.04 (2.77)		0.16	13.36
ROA	Yes	0.03 (6.23)	0.02 (4.99)	−0.00 (−0.07)	0.00 (4.01)	−0.11 (−13.24)	0.03 (3.45)		0.33	33.44

Note: Including both SHSE and SZSE samples, 1993–95. Number of observations = 673.

tercept. The results are given in Table 13. There appears to be a quadratic relation between the market-to-book value ratio and legal person ownership for both stock exchanges in 1993 through 1995. It is worth noting, however, that the signs of estimated coefficients with FLP and FLP² indicate a U-shaped rather than hump-shaped curve. The value of firms decreases with FLP when FLP is low, but increases when it is high. But this effect does not exist for ROE and ROA regressions. When running separate regressions, the relation between MBR and legal person ownership remains U-shaped for the SHSE sample companies, with a minimum occurring at FLP = 0.33 in 1994 and at 0.32 for 1995. For the SZSE 1994 sample, the squared term is significant, but the linear term is not.

The U-shaped function seems to be consistent with the hypothesis of Morck et al. (1988). When legal persons own a small stake in a company, they may try to exert their influence on or collude with the management for undertaking business operations or investments that will benefit themselves but harm the firm's value in the long run. When their equity holding in the firm increases, their goal coincides with that of outside shareholders, that is, to maximize the firm's value. The market value of the firm decreases first with legal person ownership because investors see the conflicts of interest and then increases when outside shareholders anticipate the convergence of interests at a high level of legal person holdings.

The control mechanism that the legal persons rely on is not the threat of a takeover, which underlies the empirical work of McConnell and Servaes (1990). Instead, being fired by the board of directors seems to be a far more serious threat to the Chinese managers than an outsider takeover. In fact, active takeover markets do not exist in China because state shares and legal person shares are not allowed to be traded at the two stock exchanges. Even though state and legal person shares are transferable, parties involved must go through a tedious procedure and have their deals approved by the BSPM, CSRC, and the local government. In 1994 and 1995, only 32 state or legal person share transactions were registered with the two stock exchanges.¹⁷ Most of the transactions appear to have been conducted for reasons other than reorganization.¹⁸ In only two of the 32 cases, shares transferred exceed 50% of the total stock.

Legal person owners ensure managers work in the interest of shareholders through direct con-

¹⁷ *Securities Market Herald*, October 1995.

¹⁸ It seems that the transfers were motivated mainly by cashing out, gaining an access to the capital market, and diversification or consolidation.

trol. Sitting on the board with a substantial portion of shares, large legal person shareholders are able to change the management team. Legal person owners do not have to have 50% of shareholders to vote with them to replace the incumbent management. The number of shareholders of the publicly traded companies ranges from 3,000 to 100,000, with only approximately 100 (or 200 at most) attending the annual shareholder meetings. Suppose the 200 shareholders represent 60% of total outstanding shares. What the board needs to remove the management is two thirds of the 60%, and that is 40% of total shares. If a single legal person shareholder owns 20% of a firm, it should not be difficult for the legal person to gain another 20% from other legal person shareholders. Thus the inactivism of small shareholders makes legal persons a more powerful shareholder group in corporate affairs than their equity stake indicates.

Legal person shareholders have played a positive role in monitoring the management and improving a firm's performance. They depend on direct control from the board rather than a takeover market to force the management into value maximization. Their positions in the stock companies are probably weaker than those of German and Japanese banks (see, e.g., Gorton & Schmid (1996) and Prowse (1994)), but very likely stronger than that of American institutional investors. In the German case, Gorton and Schmid (1996) present evidence that the performance of German corporations rises with banks' equity holdings in the corporations. Smith (1996) shows that even without board seats, American institutional investors monitor firms in which they have an investment, and their contribution to the firms' market performance is positive and significant.

2.4. *The inefficiency of state ownership*

Why does the state behave differently from legal persons? Boycko et al. (1995) suggested that the government may pursue political objectives such as excess employment rather than profit maximization. Many studies also point out that conflicts between the government's and shareholders' goals are a source of inefficiency. Constrained by limited data, we are unable to test the hypothesis of conflict of interest, but we address the issue indirectly and provide further evidence.

If employment is one of the government's objectives, state-dominated companies are likely to hire more workers, and *ceteris paribus*, have lower labor productivity. Because the CSRC does not require listed companies to report the number of workers hired, it is difficult to collect employment data. The data set used in this section is obtained from 100 SHSE-listed companies for 1993 and 1994. Using this limited data set, we estimate the following equation for the 100 firms:

$$\log\left(\frac{Y_j}{L_j}\right) = \alpha_i + \beta_i \log\left(\frac{K_j}{L_j}\right) + \gamma F_j + u_j, \quad (3)$$

where $I = 1, 2, \dots, 5$ for the five industries as classified by the SHSE. Y_j is the gross profits of firm, j , before taxes. K_j stands for capital stock of the firm, j , equal to the book value of total assets. L_j represents the number of employees. The dummy variable α_i is included to reflect differences in labor productivity across industries. K_j / L_j is the capital-to-labor ratio. Although the equation looks like a production function, it should not be interpreted as one because the dependent variable is not value added per worker, but is the average profits cre-

Table 14

Ownership structure and labor productivity—Dependent variable, $\log(Y_j/L_j)$

	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\alpha}_3$	$\hat{\alpha}_4$	$\hat{\alpha}_5$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$	$\hat{\beta}_5$	$\hat{\gamma}$	R ²
1993												
FST	−2.55 (−17)	−2.93 (−10)	−2.00 (−20)	−2.31 (−11)	−2.36 (−26)	0.94 (11)	0.72 (4.3)	0.99 (39)	1.13 (9.3)	0.94 (27)	−0.37 ^a (−2.1)	0.91
FLP	−2.78 (−18)	−3.14 (−10)	−2.27 (−25)	−2.53 (−11)	−2.53 (−26)	0.95 (11)	0.72 (4.0)	0.97 (47)	1.11 (8.9)	0.97 (36)	0.34 ^a (2.0)	0.91
1994												
FST	−2.64 (−21)	−2.71 (−11)	−2.28 (−13)	−2.17 (−8.1)	−2.78 (−14)	1.06 (9.9)	0.97 (3.5)	1.02 (37)	0.97 (7.6)	1.07 (7.2)	−0.61 ^a (−2.2)	0.82
FLP	−2.92 (−22)	−2.93 (−8.9)	−2.65 (−21)	−2.37 (−8.2)	−3.04 (−15)	1.11 (10)	1.04 (3.5)	1.00 (37)	0.90 (7.2)	1.13 (6.8)	0.42 ^a (2.1)	0.82

^a Significant at the 5% level.

ated by each employee. We argue that profit per work is a better indicator for the purpose of testing whether the state pursues excess employment. If Y_j were value added, higher employment would increase Y_j and L_j simultaneously, and hence the variations of the dependent variable would be smaller. It would be more difficult to detect the correlation, if it exists, between labor productivity and state ownership. An equity fraction variable, F , enters Eq. (3) to capture the effect of certain types of ownership on labor productivity. Estimation results of are given in Table 14.

The first two lines of Table 14 report estimates of the coefficients in Eq. (3) when F represents the fraction of FST. $\hat{\gamma}_i$ is negative and differs significantly from zero at the 5% level. The negative correlation of the labor productivity with the state ownership does not contradict the hypothesis that employment is one of the state's objectives. It also reinforces the results obtained earlier, that is, state ownership has an adverse effect on the performance of firms. Lines 3 and 4 in Table 14 show the estimation of Eq. (3) when F represents the fraction of FLP. The coefficient, $\hat{\gamma}$, is positive and significant at the 5% level, suggesting the greater the stake of legal person owners, the higher the labor productivity. We then estimate Eq. (3) for the same 100 firms in 1994 and report the estimated coefficients in the second half of Table 14. The results are virtually identical to those of 1993: the labor productivity falls with state ownership, but rises with legal person stakes.

The inefficiency of state ownership may arise from conflicts of interest between the central government agency, BSPM, and other shareholders. When a company must raise new capital through rights offerings, the BSPM and its representatives on the board will vote against it because the offerings will likely dilute state shares. The blockage by the state may cost the firm investment opportunities and hurt its long-term growth. If this is true, we expect DAR to be higher in state-dominated companies, but lower for those controlled by legal person shareholders. To see if this is the case, we regress DAR on five industry dummies and two ownership fraction variables, FST and FLP. Table 15 summarizes the testing results. For the SHSE samples, the coefficients associated with FST are all positive and significant at the 1% level, and those of FLP are significant and have an negative sign as expected. The correlation between the debt-to-asset ratio and ownership structure at the SZSE is weaker, indicating a relatively weaker influence of the state on corporate policies. The evidence is consis-

Table 15

Ownership structure and debt-to-asset ratio, dependent variable: DAR

	D ₁ *	D ₂	D ₃	D ₄	D ₅	FST	FLP	R ²
1993								
SHSE	0.33 (11.1)	0.24 (7.76)	0.10 (2.63)	0.41 (8.49)	0.30 (5.44)	0.22 ^a (4.28)		0.28
	0.47 (18.5)	0.36 (9.75)	0.24 (5.57)	0.56 (9.77)	0.39 (6.73)		−0.16 ^a (−2.81)	0.22
SZSE	0.38 (9.52)	0.35 (4.98)	0.36 (2.65)	0.48 (6.67)	0.39 (5.18)	0.05 (0.56)		0.00
	0.44 (14.0)	0.43 (5.15)	0.42 (3.05)	0.55 (8.45)	0.45 (5.96)		−0.15 ^c (−1.68)	0.00
1994								
SHSE	0.33 (14.0)	0.26 (9.41)	0.19 (4.74)	0.47 (8.05)	0.43 (12.5)	0.21 ^a (4.41)		0.20
	0.42 (21.8)	0.36 (11.6)	0.30 (7.31)	0.57 (8.85)	0.51 (10.1)		−0.10 ^b (−2.16)	0.12
SZSE	0.39 (12.4)	0.44 (7.77)	0.44 (4.74)	0.52 (11.7)	0.46 (7.20)	0.08 (1.18)		0.02
	0.45 (15.3)	0.51 (7.93)	0.51 (4.85)	0.58 (11.8)	0.51 (8.60)		−0.12 ^c (−1.74)	0.04

^a Significant at the 1% level.

^b Significant at the 5% level.

^c Significant at the 10% level.

* D₁ through D₅ are the five industry dummies.

tent with the hypothesis that companies in which the state has a large stake may have to rely more heavily on debt financing because the state in general opposes rights offerings.¹⁹

3. Conclusions

The empirical evidence presented here suggests the importance of large institutional shareholders in corporate governance, the inefficiency of state ownership, and potential problems in an overly dispersed ownership structure. First, the market-oriented reform measures that China has adopted seem to have improved the economic efficiency of the state sector. However, optimal resource allocation is unlikely to be achieved simply by creating markets for products, workers, and managers without changing the ownership structure of SOEs. The internal incentive structure of SOEs must be reformed by diversifying the state ownership and by introducing other forms of large shareholders, including institution investors. It would be a Pareto improvement if the government reduces or sells off the shares it holds in the stock companies.

¹⁹ In the literature, the effects of financial leverage on a firm's value can be offsetting. It is positive with a higher debt-to-asset ratio if taking into account the value of corporate tax shields and negative according to the pecking order theory. The hypothesis proposed here about the conflicts of interest of the state and other shareholders implies that the value of firms is inversely related to financial leverage. The overall effect of the debt-to-asset ratio depends on which of the factors dominate the opposing ones.

Second, if ownership diversification is needed, is dispersed private ownership then the answer for large and medium sized enterprises in China? Our results seem to suggest a negative answer. The evidence shows that the influence of individual shareholders on a firm's profitability is insignificant, if not completely irrelevant. In many cases, the coefficients for the fraction of equity held by individual investors are significant but *negative*, indicating that the market values individual private ownership in a downward manner. Apparently these publicly traded corporations in China experience the traditional free-ride problem. Individual shareholders have no incentive and no capability to monitor and influence the behavior of the management. Therefore a certain degree of ownership concentration is needed.

Third, we found a positive correlation between ownership concentration and a firm's performance. In particular, a firm's profitability is positively and significantly correlated with the fraction of legal person shares, suggesting that large legal person shareholders (institutional investors) have the incentive and the power to monitor and control the behavior of the management and have played a significant role in corporate governance. The result is robust when indicators of both concentration and ownership mix are included in the regressions. The results are largely consistent with those of Claessens et al. (1996).

Comparative studies show that in OECD countries, ownership and control rights are increasingly concentrated in the hands of financial and nonfinancial institutions (see Annex of Xu & Wang, 1997). The driving force behind this trend seems to be related to the benefit of ownership concentration as a direct measure of corporate control and other factors. Looking at the determinants of corporate control mechanisms, many studies argue convincingly that the role of large institutions in corporate governance is particularly important in countries where the legal protection of minority shareholder's interest is weak for historical and institutional reasons—a situation that in China and many other transitional economies.

Caution is needed when interpreting the results. First, this study uses a subsample of Chinese enterprises: a clean and perhaps better-performed group of enterprises that were chosen to be listed on the two stock exchanges. Our empirical study suffers unavoidably from a sample selection bias. Therefore the results apply only to large and medium sized corporations. Second, it is suggested that there could be endogeneity of ownership in transition economies; and that is, institution owners can choose to buy shares in better performing firms and leave all poorly performing ones in the hands of the state. We argue that this is not likely in China, where the state has the control over which company is to be listed and how many shares remain in the hands of the state. Unlike the case in Eastern and Central Europe, legal person owners in China have less power to select than the state. If there had been endogeneity of ownership, it would be inconceivable that the state would choose to keep the shares in poorly performing firms (as implied by our results). Nonetheless, this issue and many others must be tested in future studies. This study could also be complemented by more general studies using random samples of Chinese enterprises with quantifiable indicators of ownership mix, which is the intention of several ongoing research efforts of which we are aware.

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Appendix

The data set includes all SHSE- and SZSE-listed companies for 1993, 1994, and 1995. Major sources of information are listed as follows.

1. Publications by CSRC, the SHSE and SZSE, and China's Securities Association such as: the China Securities Annual Report, 1994, 1995, and 1996; SHSE Securities Yearbook, 1993, 1994, and 1995; and SZSE Fact Book, 1993, 1994, and 1995. Most of the figures on ownership mix and accounting ratios used in the regressions are gathered and calculated from these publications.
2. The Information and Statistic Department of the two exchanges, which kindly provided us daily trading data, including open and closing prices, volume, and value.
3. 1995 annual reports of listed companies. We collected annual reports of more than 100 SHSE companies and for about 60 SZSE companies. We relied on these reports for information on the board and the supervisory committee and the top 10 shareholders. Using the information, we were able to study the composition of the board and the committee and to compute ownership concentration ratios.
4. On-site study of six stock companies. However, interview data is not used in this paper.

Definition of variables

MBR the market-to-book-value ratio, share prices on the last trading day of each year multiplied by the number of total outstanding shares divided by the book value of equity

ROE return on equity after tax profits divided by the book value of equity

ROA return on assets after tax profits divided by the book value of total assets

FST the fraction of equity owned by the state, or the number of shares held by the state divided by the number of total outstanding shares

FLP, FTA, FEM, and FBS calculated similarly for the fraction of equity owned by legal persons, tradable A-share investors, employees, and B-shareholders, respectively.

A5 a concentration ratio, percentage of shares controlled by top five shareholders.

A10 a concentration ratio, percentage of shares controlled by top 10 shareholders.

HERF Herfindahl index of ownership concentration, the sum of squared percentage of shares controlled by each top 10 shareholder.

DUM_{*i*}, $I = 1, 2, \dots, 5$, for manufacturing, trade, utility, real estate industries, and conglomerates, respectively. These industry dummies are set according to the classification of listed corporations by the two stock exchanges. Let d_{ij} be the element of DUM_{*i*}, $d_{ij} = 1$ if firm j is in industry I and $d_{ij} = 0$ otherwise.

SALE operating sales in billion Renminbi (RMB in short, the unit of Chinese currency). It measures the size effect of firms.

DAR the debt-to-asset ratio, which equals the book value of debt divided by the book value of assets. In the United States, debt financing has a tax advantage over equity financing, and hence the market value of a firm with a greater leverage is expected to be higher. The direction of DAR's effect on the Chinese stock companies, however, is unknown at this point.

GROW growth of net income. Stocks of companies with high growth should be priced higher in an efficient market.

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